Lab Sheet-03

Object Oriented Programming Using Java

LastDate : 12-10-2026

MCA III Semester

Part A – 1-D Arrays (Q1–Q6)

1. Write a Java program to input and display 5 integers using a 1-D array.
2. Create a Java program to input 10 integers and display them in reverse order.
3. Write a Java program to find the sum of all elements in a 1-D array.
4. Create a Java program to find the maximum element in an array.
5. Write a Java program to count even and odd numbers in an array.
6. Create a Java program to copy all elements from one array to another.

Part B – 2-D Arrays (Q7–Q13)

7. Write a Java program to input and display elements of a 3 × 3 matrix.
8. Create a Java program to find the sum of all elements in a matrix.
9. Write a Java program to find the sum of each row in a matrix.
10. Create a Java program to find the sum of each column in a matrix.
11. Write a Java program to transpose a matrix.
12. Create a Java program to add two matrices of the same size.
13. Write a Java program to multiply two matrices (basic case with 3 × 3 matrices).

Part C – 3-D Arrays (Q14–Q18)

14. Write a Java program to input and display elements of a 2 × 2 × 2 3-D array.
15. Create a Java program to find the sum of all elements in a 3-D array.
16. Write a Java program to count positive and negative numbers in a 3-D array.
17. Create a Java program to find the maximum element in a 3-D array.
18. Write a Java program to search for an element in a 3-D array and display its position.

Part D – Jagged Arrays (Q19–Q20)

19. Write a Java program to create and display a jagged array with 3 rows of different lengths.
20. Create a Java program to store marks for students where each student has taken a different number of subjects, then display them.

Part E - Additional Questions

21. Write a program to rotate an array to the left by one position.
22. Count how many times each element appears in a 1-D array.
23. Find the second largest element in a 1-D array.
24. Remove duplicate elements from a 1-D array (without using collections).
25. Find the sum of the main diagonal and secondary diagonal elements of a square matrix.
26. Display the upper and lower triangular parts of a square matrix separately.
27. Print only the boundary elements of a matrix.
28. Search for an element in a 2-D array and display its row & column position.
29. Sort the elements of each row in a jagged array.
30. Check if a given matrix is a sparse matrix (more zero elements than non-zero).